CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (AOC) Examination March 2018

OC 823 Chemistry of Heterocycles

Date: 27/03/2018 Day: Tuesday Time: 01.30 pm To 02.00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I [A] Choose the correct answer for the following questions. [10]

1 Electrophilic aromatic substitution reaction of pyridine occurs at……….

a) C_2 position b) C_3 position

c) C_4 position d) C_2 & C_4 position

2 3-Hydroxypyridine mainly exists as …..

a) both keto and enol in equal amount

b) keto form

c) enol form

d) major in keto and minor in enol

3 Nucleophilic substitution reaction of pyridine is not straight forward because…..

a) presence of freely available lone pair on nitrogen atom.

b) last step involves proton transfer which required reducing agent.

c) last step involves hydride transfer which requires oxidizing agent.

d) none of above

4 Electrophilic aromatic substitution of pyridine is extremely difficult due to…..

a) lone pair on nitrogen is not a part of aromatic sextet.

b) presence of nitrogen.

c) lone pair on nitrogen is a part of aromatic sextet.

d) none of above

5 What is the basic difference between Conrad – Limpach reaction and Knorr modification?

a) Conrad – Limpach reaction proceeds via acid catalyzed reaction while Knorr modification proceeds via base catalyzed reaction.

b) Conrad – Limpach reaction carried out at room temperature while Knorr modification carried out at higher temperature.

c) Conrad – Limpach reaction carried out at higher temperature while Knorr modification carried out at room temperature.

d) Both reaction carried out in different solvent

6 Which salt of pyrylium is most stable?

a) hexaflouro phosphate b) perchlorate

c) tetraflouro borate d) chloride

7 Indole upon bromination in presence of water gives

a) 3-hydroxy indole b) 2- hydroxy indole

c) 2-bromo indole d) 3-bromo indole

8 The role $FeSO_4$ in skrup synthesis is ………

a) to control vigorous condition. b) to act as an oxidizing agent.

c) to act as a acid catalyst. d) to increase the yield of the product.

9 The reactivity order of displacement reaction of substituted pyridine is ……..

a) 4 > 2 > 3 b) 3 > 2 > 4

c) 2 > 4 > 3 d) 4 > 3 > 2

10 The resonance energy of pyridine is ______ kcal/mole.

a) 36 b) 27.9

c) 29.7 d) 34.9

Q – I [B] Give the name of following heterocyclic compounds. [10]

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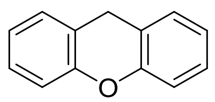

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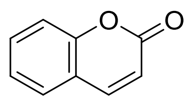

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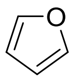

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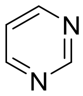

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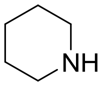

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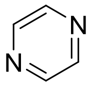

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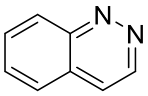

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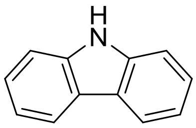

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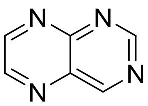

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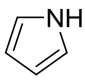


CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (AOC) Examination March 2018

OC 823 Chemistry of Heterocycles

Date: 27/03/2018 Day: Tuesday Time: 02.00 pm To 04.30 pm Maximum Marks: 50

Important Instructions:

- **Section I and II must be attempted in Two Separate ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION -I **[20]**

Q - II

[A] Write a note on Hantzsch Pyridine synthesis. **[04]**

[B] What compounds would result from the following reagent combination: **[03]**

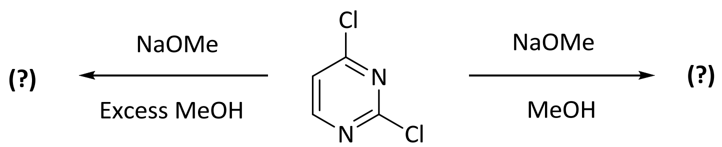
i) Cyanoacetamide with acetyl acetone

ii) Benzaldehyde with Acetyl acetone and NH_3

[C] "Pyridine upon nitration gives 3-nitropyridine but not 2-and 4-nitropyridine." Explain. **[03]**

OR

[C] Predict the products of following reaction and give comment on formation of products. **[03]**



Q-III

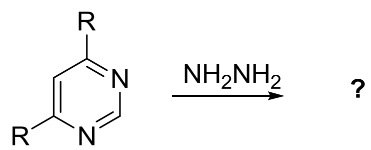
[A] Write a note on Guareschi synthesis. **[04]**

[B] Write the synthesis of substituted pyrimidine starting from **[03]**

i) Ethylacetoacetate and Guanidine

ii) Acetyl acetone and Acetamidine

[C] Complete the following reaction by giving suitable mechanism. **[03]**



OR

[C] Deduce the structure for the products formed in turn by reacting pyridine with, $NaNH_2$, 100˚C to give **A** ($C_5H_7N_2$), this with Br_2 to give **B** ($C_5H_6N_2Br$), then **B** is reacted with HNO_3/H_2SO_4 to produce **C** ($C_5H_5N_3BrO_2$). **[03]**

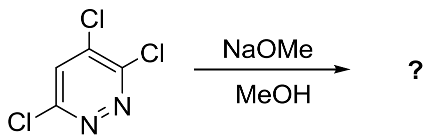
SECTION -II **[30]**

Q-IV

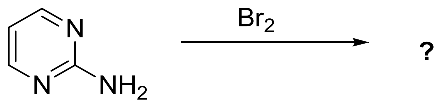
[A] Write a note on Skraup Quinoline synthesis. **[04]**

[B] Complete and rewrite the following reactions. **[03]**

i)



ii)



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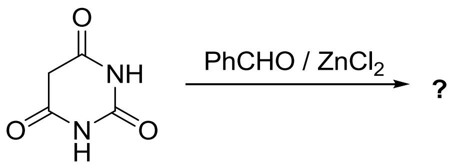
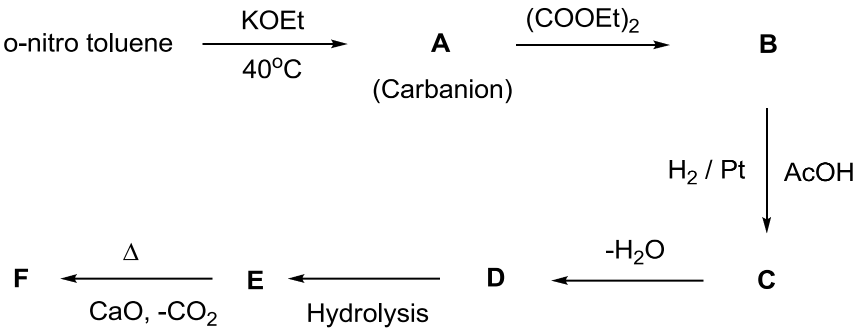


[C] Write the mechanism of Vilsmeier-Haack reaction of indole. **[03]**

OR

[C] Complete the following reaction path. **[03]**



Q-V

[A] Write a note on Fischer Indole Synthesis. **[04]**

[B] What compounds are produced at each stage in the following sequence: **[03]**

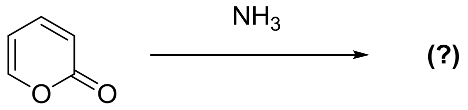
i) 2,4,6-trimethyl pyrimidine reacted with PhLi followed by Methyl Iodide

ii) 2,4,6-trimethyl pyrimidine reacted with PhLi followed by Ethylbenzoate

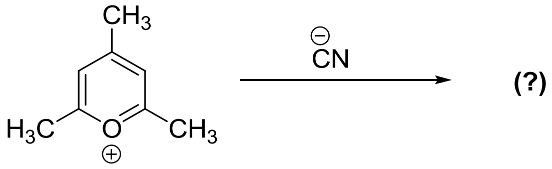
iii) 2,4,6-trimethyl pyrimidine reacted with PhLi followed by Ph_2S_2.

[C] Complete and rewrite following reactions with its mechanism. **[03]**

i)



ii)



OR

[C] Complete the following reaction by giving appropriate mechanism for the formation of product. **[03]**

$$\beta\text{–phenylethylamine} \xrightarrow[\text{(ii) POCl}_3]{\text{(i) CH}_3\text{COCl}} (?)$$

Q-VI

[A] Write a note on Reissert reaction of quinoline. **[04]**

[B] Complete the following reaction path. **[03]**

$$\text{Methyl Anthranalate} \xrightarrow[\text{MeMgBr}]{\text{Excess}} \textbf{A} \xrightarrow[-H_2O]{P_2O_5} \textbf{B} \xrightarrow[\text{HCl}]{\text{NaNO}_2} \textbf{C}$$

$$\textbf{C} \xrightarrow{\text{Michael Type Addition}} \textbf{D}$$

$$\underset{\text{(cinnoline)}}{\textbf{F}} \xleftarrow[\text{-CO2}]{\text{D}} \textbf{E} \xleftarrow{[O]} \textbf{D}$$

[C] Predict the product and write the reaction mechanism, when **[03]**

i) γ-pyrone reacted with phenyl hydrazine

ii) 4-methyl pyrylium salt reacted with ammonia

OR

[C] Which starting materials would be required for the Fischer-Indole synthesis of following indole derivatives? **[03]**

i) 3-methyl indole

ii) 1,2,3,4-tetrahydro carbazole

iii) 2,3-diphenyl indole